DATA ENGINEERING · PORTFOLIO PROJECT

OlympiaFlow

Olympics Data Warehouse

From two raw CSV files to a reconciled, audited and recoverable star schema, built with Informatica PowerCenter and Oracle.

271,116	6	10	2
source rows, all loaded	tables in the star schema	mappings and sessions	recovery scenarios tested

BUILT WITH

Informatica PowerCenter · Oracle · SQL Developer

INSIDE

Layered ETL · Dimensional modeling · Data reconciliation · Audit and recovery

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01 The Project

Medals are easy to count. Trusting the numbers behind them takes engineering. This project loads the Kaggle dataset *120 years of Olympic history* (athlete results from 1896 to 2016) through three Oracle layers and publishes a star schema for analysis. It follows production habits: parallel worklets, explicit dependencies, an audit trail for each session and a recovery plan tested by breaking the pipeline on purpose.

Layered
ODS, STG, DWH

Modelled
star schema, SCD 0 and 1

Reconciled
counts checked per layer

Recoverable
suspend, then recover

Questions the model answers

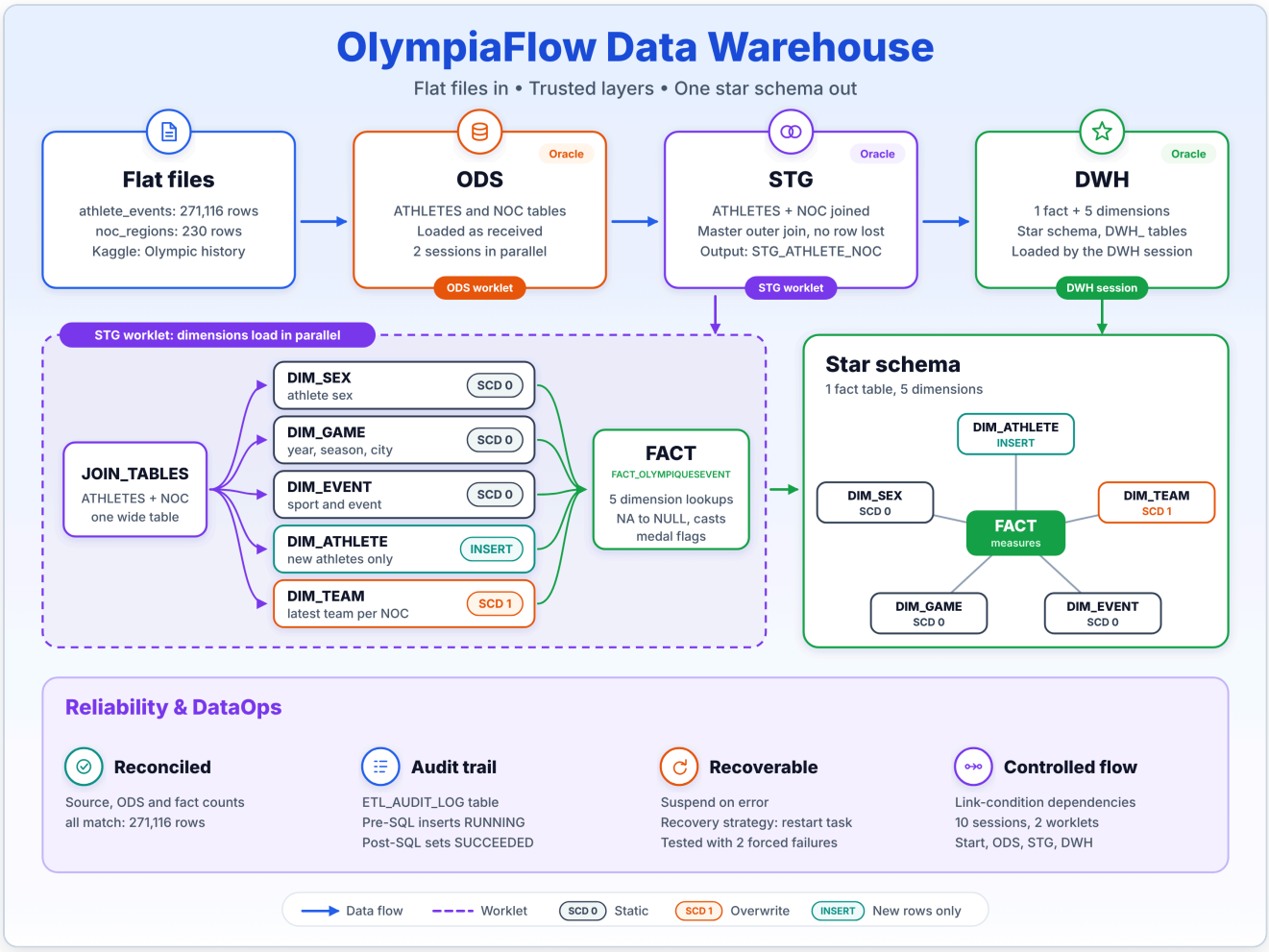
Area	Example question
Medals	How many gold, silver and bronze medals does each country win, overall and by Games?
Participation	How did the number of athletes, sports and events grow from 1896 to 2016?
Athlete profile	How do age, height and weight differ by sport, sex and medal result?
Teams	Which team label represents each country, and how has it changed?

Source files

File	Rows	Columns
athlete_events	271,116	ID, Name, Sex, Age, Height, Weight, Team, NOC, Games, Year, Season, City, Sport, Event, Medal
noc_regions	230	NOC code and region name

02 Architecture

Data is loaded as received into ODS, joined and shaped in STG, and published as a star schema in DWH. Each stage is a worklet or session in one Informatica workflow.

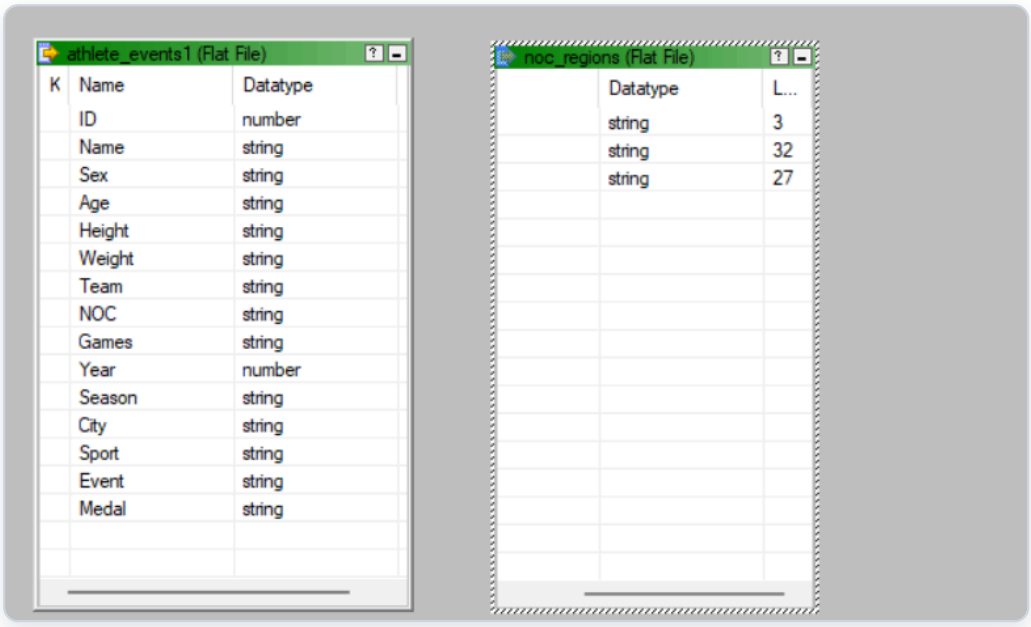


Pipeline architecture: flat files, ODS, STG, DWH, with audit and recovery

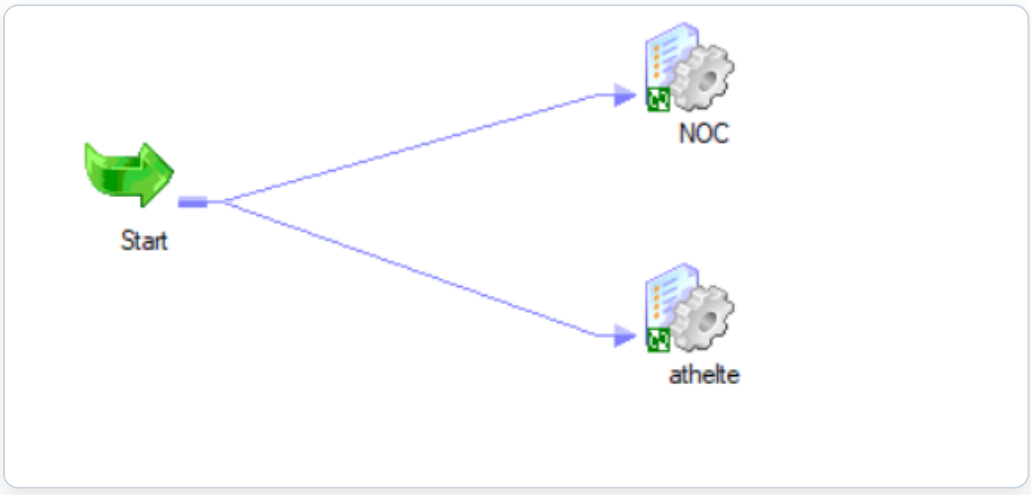
Layer	Responsibility	Built as
Flat files	athlete_events and noc_regions as CSV sources	2 files
ODS	Both files loaded as received into ATHLETES and NOC	ODS worklet, 2 sessions in parallel
STG	Tables joined into STG_ATHLETE_NOC, then 5 dimensions and the fact	STG worklet
DWH	Star schema published as DWH_DIM_* and DWH_FACT_OLYMPIQUESEVENT	DWH session

03 Source Data and ODS

Both files are defined as flat-file sources, then loaded into Oracle exactly as they arrive. The ODS worklet runs the two sessions in parallel because they do not depend on each other.



Source definitions for the flat files in Designer



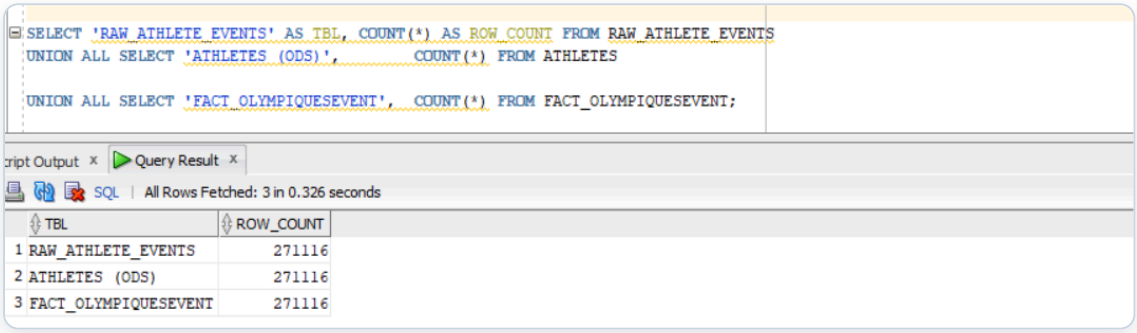
ODS worklet: the NOC and ATHLETES sessions run in parallel

Design choice. ODS keeps the data untouched. Cleaning and joining happen in STG, so the original is always available to compare against.

04 Data Quality

Comparing loaded row counts with the raw source exposed two silent data-loss problems. Both were found, explained and fixed.

Problem	What happened	Fix
Rows rejected at ODS	First load returned 206,165 rows instead of 271,116. Age, Height and Weight were typed as numbers, but the source uses the text NA (64,951 rows have it in at least one column). Informatica rejected the whole row.	Source Qualifier reads these columns as text. The fact Expression converts NA to NULL and casts the numbers.
Rows dropped at STG	Country codes differ between files (SGP versus SIN), so 349 athlete rows fell out of the join.	Master outer join with NOC as master and ATHLETES as detail. No athlete row is lost.

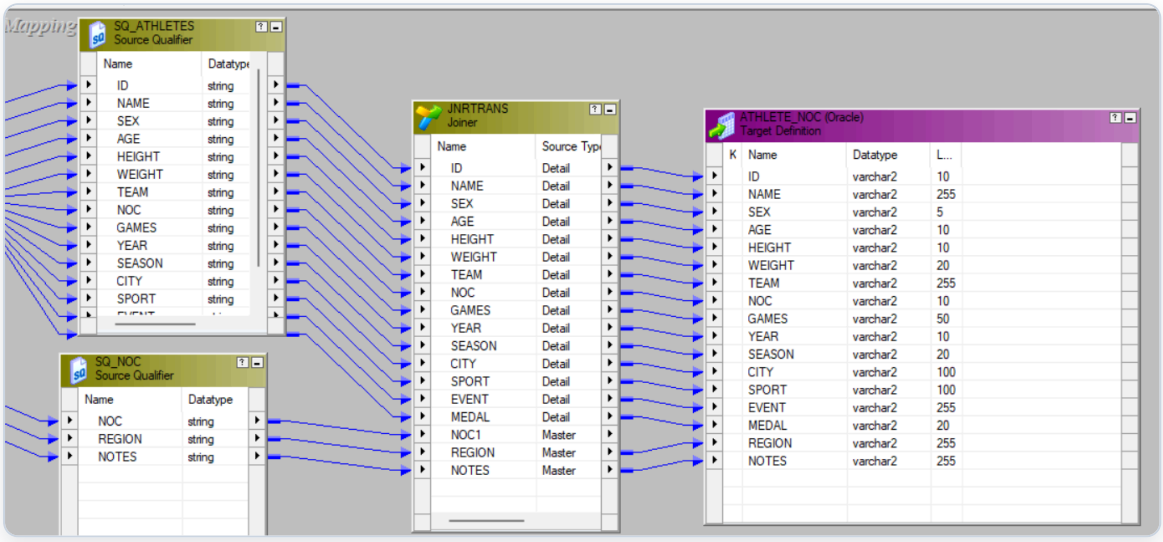


Final check in Oracle: raw 271,116, ODS 271,116, fact 271,116

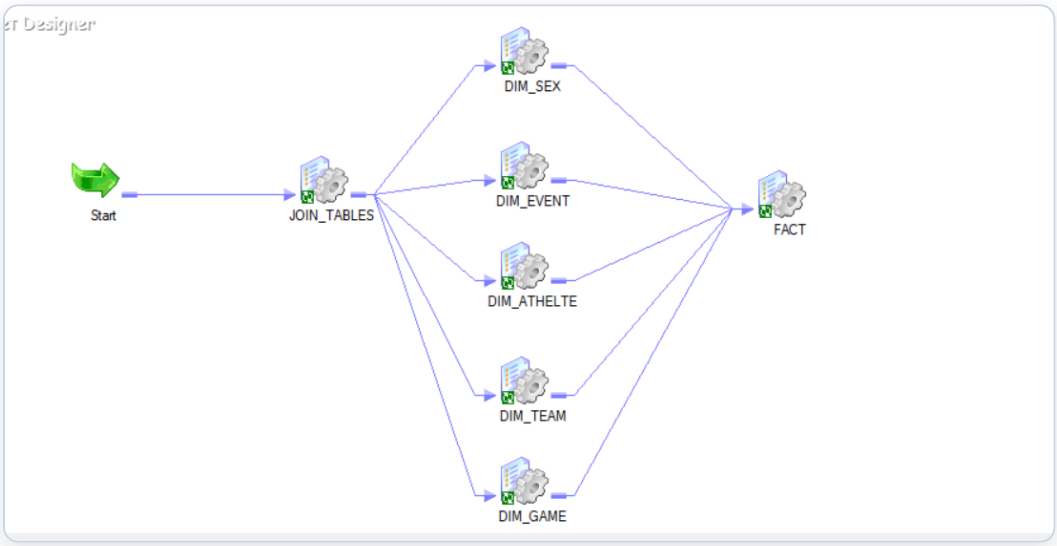
Lesson. A load that finishes without errors is not proof that it loaded everything. Row-count reconciliation catches what the logs do not.

05 STG Layer

The STG worklet starts with JOIN_TABLES, which merges athletes and countries into one wide table. The five dimensions then load in parallel, and the fact table runs only after all of them succeed.



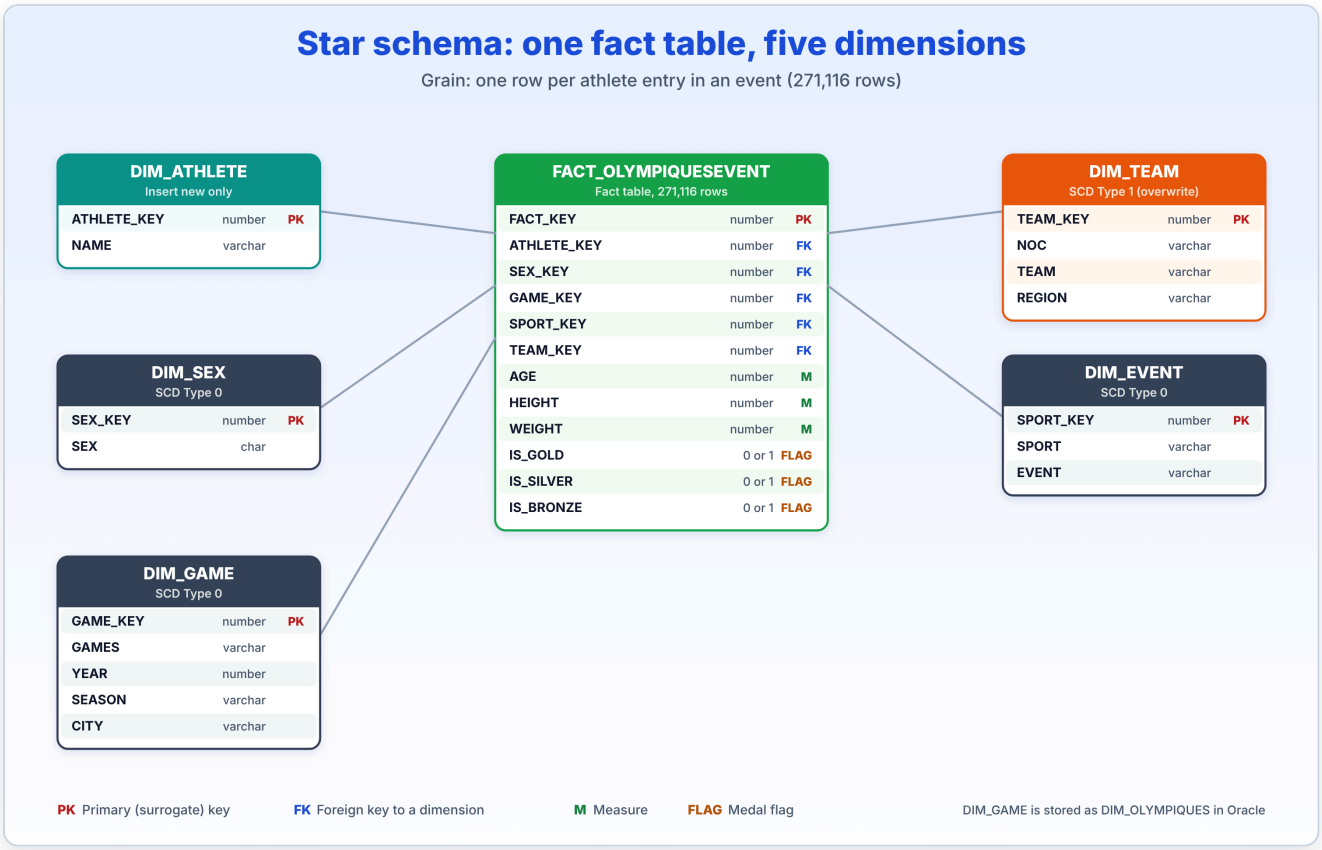
JOINING_TABLES mapping: a Joiner with NOC as master (outer join)



STG worklet: JOIN_TABLES, five parallel dimensions, then FACT

06 Dimensional Modeling

Grain: one row per athlete entry in an event, so the fact table holds exactly the source row count.
Measures: age, height, weight and three medal flags.

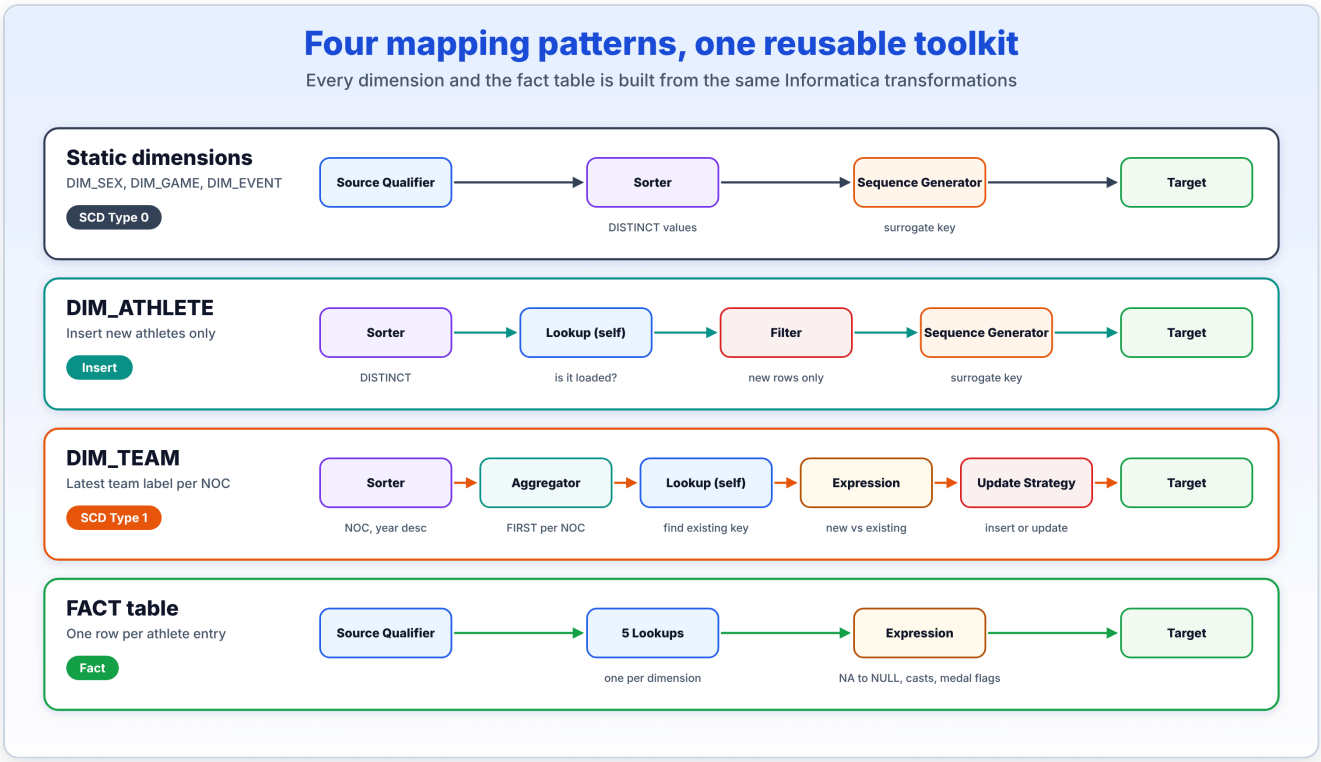


Star schema with columns as built in Oracle

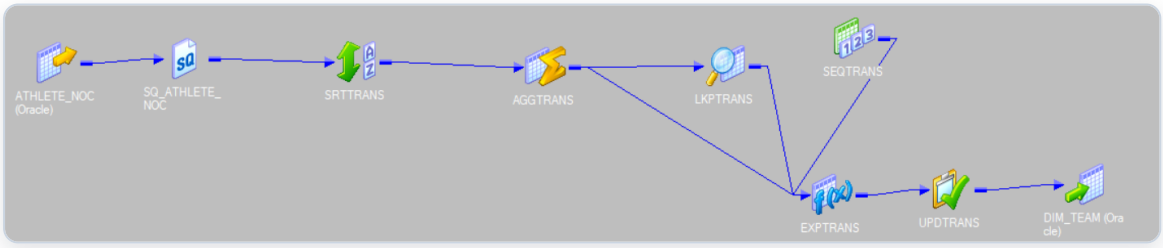
Dimension	Change handling	Reasoning
DIM_SEX, DIM_GAME, DIM_EVENT	SCD Type 0	Attributes never change for a natural key, so nothing is tracked.
DIM_ATHLETE	Insert new only	A lookup stops an athlete being inserted twice, so reruns are safe.
DIM_TEAM	SCD Type 1	A NOC can carry different team labels across years. The latest wins, no history.

07 Mapping Patterns

Ten mappings are built from a small set of transformations. Each pattern matches how the data behaves.



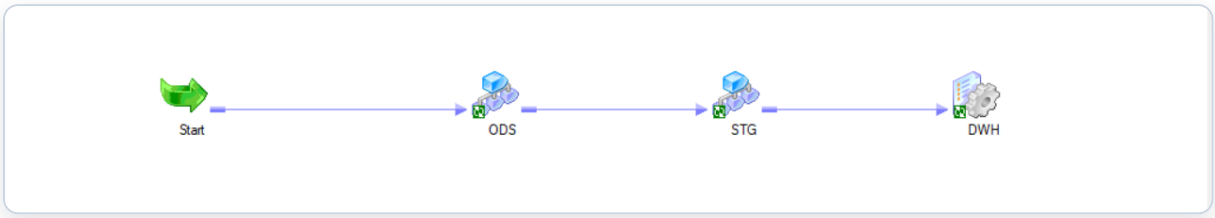
The four patterns used across dimensions and the fact table



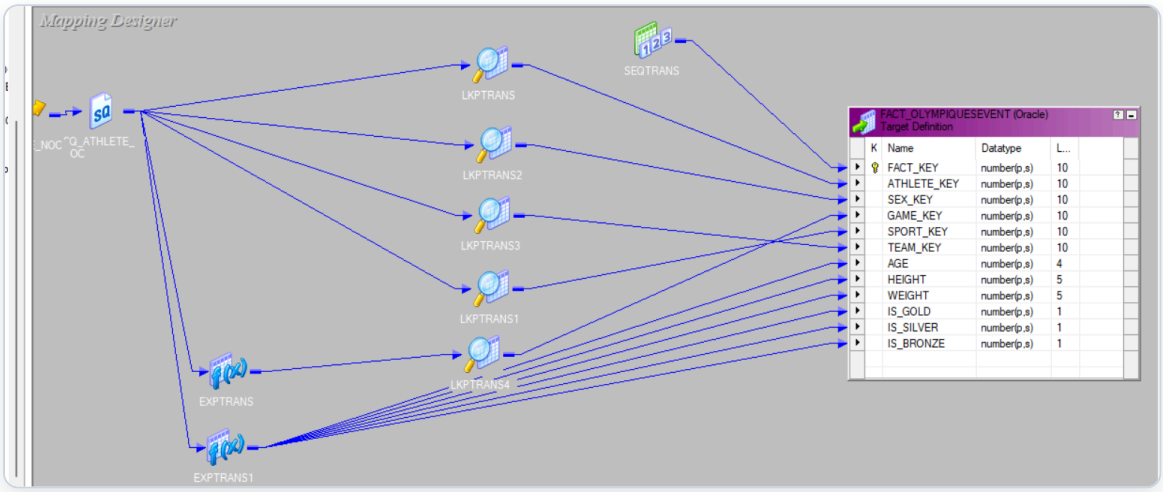
DIM_TEAM mapping: Sorter, Aggregator, Lookup, Expression, Update Strategy

08 Workflow and Orchestration

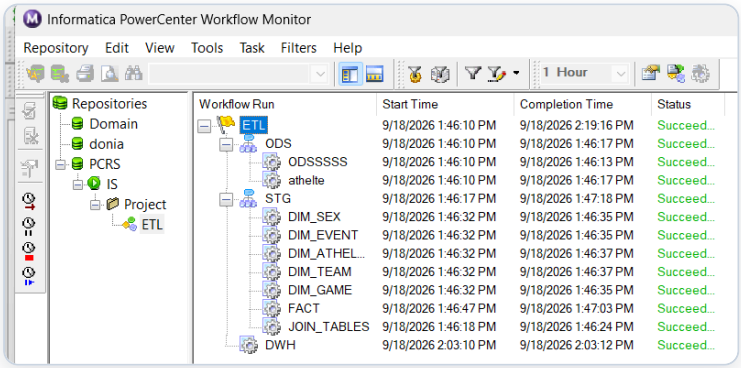
The ETL workflow runs Start, ODS, STG and DWH, connected by link conditions so that a stage starts only when the previous one succeeded.



ETL workflow: Start, ODS, STG, DWH



FACT mapping: five lookups and an Expression



Workflow Monitor: every session succeeded

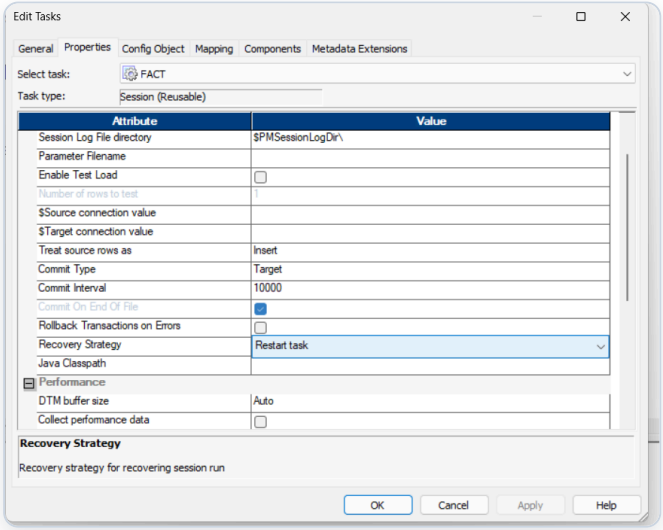
09 Reliability: Audit and Recovery

Audit trail. Each session writes to ETL_AUDIT_LOG. Pre-SQL inserts a RUNNING row, and Post-SQL updates it to SUCCEEDED. One query shows whether a run finished.

	AUDIT_ID	WORKFLOW_NAME	SESSION_NAME	START_TIME	END_TIME	STATUS
1	2	ETL	ODSSSSS	18-SEP-26	18-SEP-26	SUCCEEDED
2	1	ETL	athelte	18-SEP-26	18-SEP-26	SUCCEEDED
3	3	ETL	JOIN_TABLES	18-SEP-26	18-SEP-26	SUCCEEDED
4	5	ETL	DIM_GAME	18-SEP-26	18-SEP-26	SUCCEEDED
5	4	ETL	DIM_SEX	18-SEP-26	18-SEP-26	SUCCEEDED
6	8	ETL	DIM_TEAM	18-SEP-26	18-SEP-26	SUCCEEDED
7	7	ETL	DIM_ATHELTE	18-SEP-26	18-SEP-26	SUCCEEDED
8	6	ETL	DIM_EVENT	18-SEP-26	18-SEP-26	SUCCEEDED
9	9	ETL	FACT	18-SEP-26	18-SEP-26	SUCCEEDED
10	10	ETL	athelte	18-SEP-26	18-SEP-26	SUCCEEDED
11	11	ETL	ODSSSSS	18-SEP-26	18-SEP-26	SUCCEEDED
12	12	ETL	JOIN_TABLES	18-SEP-26	18-SEP-26	SUCCEEDED
13	15	ETL	DIM_SEX	18-SEP-26	18-SEP-26	SUCCEEDED
14	14	ETL	DIM_GAME	18-SEP-26	18-SEP-26	SUCCEEDED
15	17	ETL	DIM_TEAM	18-SEP-26	18-SEP-26	SUCCEEDED
16	13	ETL	DIM_EVENT	18-SEP-26	18-SEP-26	SUCCEEDED
17	16	ETL	DIM_ATHELTE	18-SEP-26	18-SEP-26	SUCCEEDED
18	18	ETL	FACT	18-SEP-26	18-SEP-26	SUCCEEDED
19	19	ETL	DWH	18-SEP-26	18-SEP-26	SUCCEEDED
20	20	ETL	DWH	18-SEP-26	18-SEP-26	SUCCEEDED
21	21	ETL	DWH	18-SEP-26	18-SEP-26	SUCCEEDED
22	22	ETL	DWH	18-SEP-26	18-SEP-26	SUCCEEDED
23	24	ETL	DWH	18-SEP-26	18-SEP-26	SUCCEEDED
24	23	ETL	DWH	18-SEP-26	18-SEP-26	SUCCEEDED

ETL_AUDIT_LOG: workflow, session, start, end and status

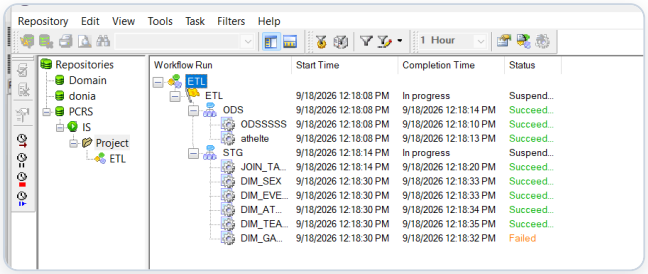
Recovery. Every session is set to suspend on error with the strategy Restart task. A failure pauses the run instead of pushing incomplete data downstream. In Workflow Monitor, Recover resumes from the point of failure without re-running finished tasks.



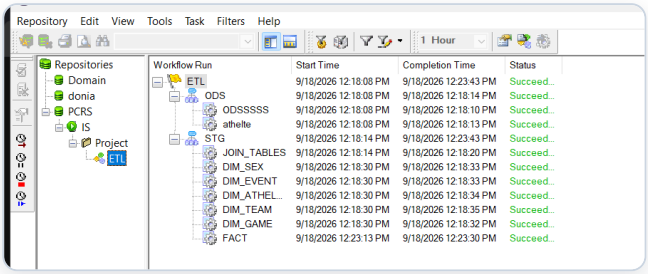
Session property: Recovery Strategy set to Restart task

10 Recovery Tests and Takeaways

Two forced failures proved the plan. The first aborted a dimension session. The second deleted a target table, which was recreated before recovery.



Workflow suspended after the failure



All tasks succeeded after Recover

Skill	Demonstrated by
ETL development	Informatica mappings, sessions, worklets, link conditions
Data modeling	Star schema, surrogate keys, SCD Type 0 and 1
Data quality	Row-count reconciliation, NA handling, outer-join fix
Reliability	Audit table, suspend on error, restart-task recovery

What's next

- Power BI dashboard on top of the star schema (medals, participation, athlete profile).
- Export mappings and DDL into the repository for version control.

Explore the project: github.com/donia444/olympiaflow